PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – III

FLEXURAL AND SHEAR

Syllabus : Analysis of beams for flexure and shear – beams pre-stressed with straight, concentric, eccentric, bent and parabolic tendons – kern line – cable profile – design of PSC beams (rectangular and I sections) using IS 1343 – analysis and design of rectangular and I beams for shear – introduction to transmission length and end block (no design and analytical problems).

Contents

1. Basic assumptions, notation, allowable stresses
2. Concentric tendon
3. Eccentric straight tendon
4. Bent & parabolic tendons – equivalent loads
5. Kern line & pressure line
6. Cable profile / cable zone
7. Design of rectangular beam (IS 1343)
8. Design of I-section beam
9. Shear in PSC beams – principal stresses, IS 1343
10. Shear design examples
11. Transmission length & end block (introduction)
12. Important questions & formula sheet

Exam tip : Design problems carry 10–15 marks. Follow the steps : Z_b → select section → P and e → check stresses at transfer and service.

1. Analysis – Basic Assumptions

1. Concrete is a **homogeneous, elastic** material – plane sections remain plane (Bernoulli).
2. The section is **un-cracked** under working loads (Type 1 / 2) – gross section properties A, I, Z used (or transformed).
3. Steel (tendon) force P is applied at the tendon centroid at eccentricity e from the C.G. of concrete section.
4. Stresses are found by **superposition** of (i) direct stress P/A, (ii) bending due to prestress $P{\cdot}e$, (iii) bending due to loads M.
5. Losses are accounted by $\eta = P_e/P_i$ (0.75 – 0.85).

Notation

Symbol	Meaning
P_i, P_0, P_e	initial, at transfer, effective prestress force
A, I	area & M.I. of the section
y_t, y_b	distance of top / bottom fibre from C.G.
Z_t, Z_b	section moduli I/y_t, I/y_b
e	eccentricity of tendon below C.G.
M_g, M_q, M_T	moments : self weight, imposed, total
r	radius of gyration, $r^2 = I/A$

General stress equation (compression +ve)

$$f = P/A \pm P\,e\,y/I \pm M\,y/I$$

$$\text{Top : } f_t = P/A - Pe/Z_t + M/Z_t \quad \text{Bottom : } f_b = P/A + Pe/Z_b - M/Z_b$$

Permissible stresses (IS 1343)

- At **transfer** : compression ≤ 0.54 f_{ci} (flexure); tension = 0 for Type 1 (small tension permitted in Type 2 as per code table).
- At **service** : compression ≤ 0.41 f_{ck} (flexure); tension = 0 (Type 1), limited tension (Type 2 – see IS 1343 Table 5).
- f_{ci} = cube strength at transfer. Minimum grade : M40 pre-tension, M30 post-tension.
- In design problems the permissible values are usually **given**; use them as given.

2. Beam with Concentric (Axial) Tendon

The tendon passes through the C.G. of the section: e = 0. Prestress produces a **uniform compression** P/A – no camber.

$$f = P/A \pm M y / I$$

Example 1

A rectangular beam 200 mm × 400 mm, span 6 m, carries total UDL 12 kN/m (including self-weight). Concentric P = 600 kN. Find extreme stresses at mid-span.

$$A = 80000\ mm^2 ;\ Z = 200 \times 400^2/6 = 5.333 \times 10^6\ mm^3 ;\ M = 12 \times 6^2/8 = 54\ kN{\cdot}m$$

$$P/A = 7.5\ N/mm^2 ;\ M/Z = 54\times10^6 / 5.333\times10^6 = 10.125\ N/mm^2$$

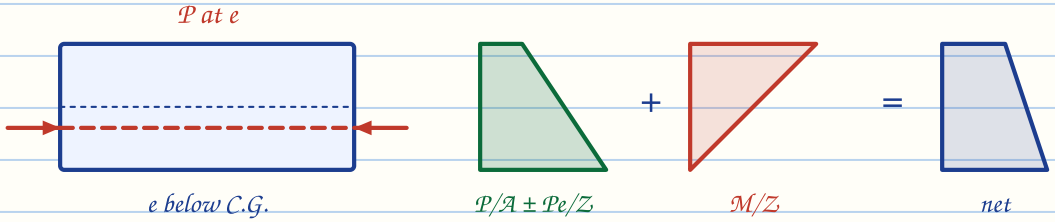
$$Top = 7.5 + 10.125 = 17.63\ N/mm^2\ (comp.) ;\ Bottom = 7.5 - 10.125 = -2.63\ N/mm^2\ (tension)$$

Net tension at bottom → cracking. Hence give an **eccentricity** – next page. Concentric prestress is used in axially loaded members (piles, poles, pipes, tension ties).

3. Beam with Eccentric Straight Tendon

Tendon is at a constant eccentricity e **below** the C.G. – the prestress now has a hogging moment P·e which **counteracts the sagging load moment**.

$$f_t = P/A - P\,e/Z_t + M/Z_t \qquad f_b = P/A + P\,e/Z_b - M/Z_b$$



Top fibre : compression from M, tension from Pe → balanced; bottom : Pe compression – M tension

Stress is generally compression throughout when e is chosen well

Fig. Eccentric prestress – stress diagrams

Example 2 (same beam, e = 60 mm)

$$P\,e/Z = 600\times10^3 \times 60 / 5.333\times10^6 = 6.75\ N/mm^2$$

$$Top = 7.5 - 6.75 + 10.125 = 10.88\ N/mm^2 ;\ Bottom = 7.5 + 6.75 - 10.125 = 4.13\ N/mm^2$$

Both fibres are in compression – no tension. **Check at transfer (self-weight only, 2 kN/m → M_g = 9 kN·m, M_g/Z = 1.69)** : top = 7.5 – 6.75 + 1.69 = 2.44 ✓ (no tension).

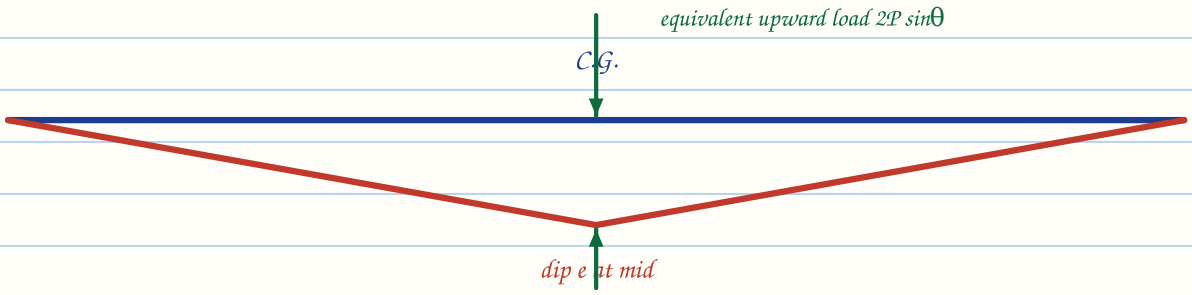
At the ends (M = 0) an eccentric straight tendon gives top tension $P\,e/Z_t - P/A$ (here 6.75 – 7.5 = –0.75 → still compressive) – if e is large, tension appears at the **top of the end sections**; remedy : (i) reduce e near the supports (bent/harped or parabolic tendon), (ii) de-bond some strands, (iii) provide top steel.

Eccentric straight tendon : simple, used in pre-tensioned beams, slabs. For longer spans use harped / parabolic tendons.

4. Bent (Harped) and Parabolic Tendons

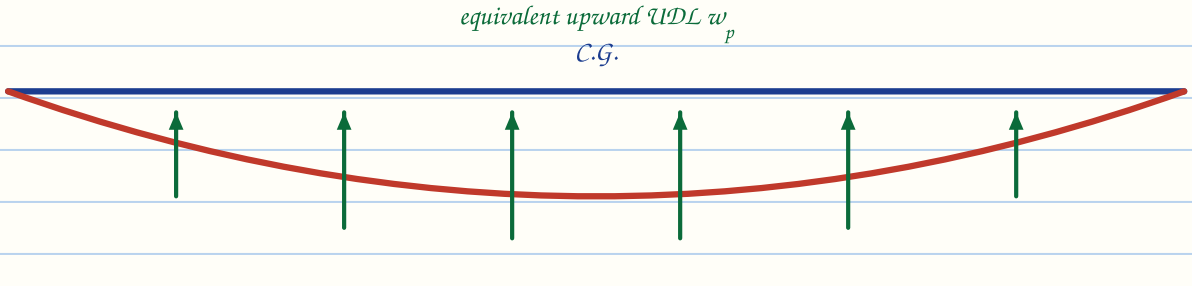
Tendon profile is varied so that e is large at mid-span (max moment) and small (zero) at supports – lesser end stresses, **vertical component of prestress reduces shear**.

(a) Bent (harped) tendon – kink at mid-span or third points



- Angle of the tendon $\theta \approx \tan\theta = 2e/L$ (end at C.G., e at midspan).
- Equivalent **upward concentrated load at the kink** $W = 2P \sin\theta \approx 2P(2e/L) = \mathbf{4Pe/L}$.
- Moment at mid-span from prestress = P·e (sagging comp. – hogging); **vertical shear component** $V_p = P \sin\theta$.

(b) Parabolic tendon – equivalent UDL



$$y = 4ex(L-x)/L^2;\ d^2y/dx^2 = -8e/L^2 \rightarrow w_p = 8Pe/L^2 \text{ (upward)}$$

- Total upward force = $w_p L = 8Pe/L$ = twice the end slope force : slope at end $4e/L \rightarrow$ vertical component $4Pe/L$ each end ✓.
- Net moment from prestress at section = P·y (y = tendon dip at that section).
- Net downward load = $w - w_p \rightarrow$ if $w = w_p$ the beam is perfectly balanced (load balancing).

5. Equivalent Loads – Summary

Tendon profile	Equivalent load	Moment due to P
Straight, concentric	No transverse load; axial P only	$M = 0$
Straight, eccentric e	End moments P·e (hogging)	$M = -P\,e$ (constant)
Bent (single kink, dip e)	Upward point load $4Pe/L$ at mid; end axial P	$M = -P\cdot y(x)$
Parabolic, dip e	Upward UDL $8Pe/L^2$; end reactions	$M = -P\cdot y(x)$
Circular arc	Radial load P/R per unit length	$M = -P\cdot y$

Example 3 (parabolic tendon, load balancing)

Beam of Example 1 (6 m, 200 × 400) with a parabolic tendon, e = 0 at ends and e = 100 mm at mid, P = 600 kN, total load 12 kN/m. Find stresses at mid-span and the net load.

$$w_p = 8 \times 600 \times 0.1/6^2 = 13.33 \text{ kN/m (upward)} > 12 \text{ kN/m} \rightarrow \text{net upward } 1.33 \text{ kN/m}$$

$$\text{Top} = 7.5 - 600000\times100/5.333\times10^6 + 10.125 = 7.5 - 11.25 + 10.125 = 6.38 \text{ N/mm}^2$$

$$\text{Bottom} = 7.5 + 11.25 - 10.125 = 8.63 \text{ N/mm}^2$$

Both compressive. At the support $M = 0$, $e = 0 \rightarrow$ uniform $P/A = 7.5$ N/mm² ✓ (no end tension).

Cambering : at transfer, only self-weight (say 2.4 kN/m) acts but the full upward 13.33 kN/m is present → large camber. Check top fibre tension at transfer: $M_g = 10.8$ kN·m → top = 7.5 − 11.25 + 2.03 = −1.72 N/mm² (tension) → provision of top steel or lesser e needed.

Check **two stages** always : (1) transfer – P_0 with M_g only; (2) service – η P_0 with M_T.

6. Kern Line and Pressure Line

Kern of a section

The **kern** is the region within which a compressive load (prestress) can be applied **without producing tension** anywhere in the section.

- For a load P at eccentricity e : $f = P/A\ (1 \pm e\,y/r^2)$. Tension is absent if $e\,y / r^2 \le 1$.
- Upper kern distance (above C.G.) : $k_t = Z_b / A = r^2 / y_b$.
- Lower kern distance (below C.G.) : $k_b = Z_t / A = r^2 / y_t$.

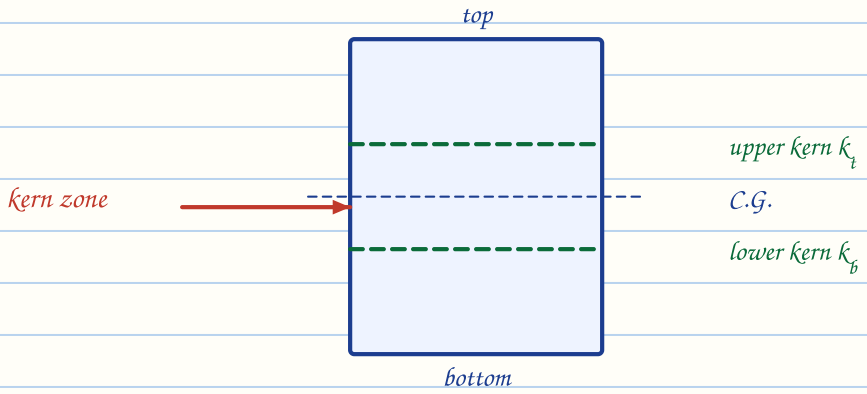


Fig. Kern zone (middle-third for rectangle)

Rectangle $b \times D$: $k_t = k_b = D/6$; I-section : $k_t = Z_b/A$, $k_b = Z_t/A$

Pressure line (thrust line, C-line)

- The locus of the **resultant compressive force** in the concrete (C-line). It is the line of action of the compressive force in concrete – for a concordant tendon C-line coincides with the cable.
- Distance a from the cable (C-line shift) : $a = M / P$ (due to external moments, C-line moves up by M/P).
- **For no tension**, the C-line must stay within the kern at all sections.

C-line shift due to load : $a = M / P$; C-line position = e – M/P (above cable)

Example 4 – Kern

Rectangular section 200 × 400. Find the kern points and check that e = 60 mm lies in the kern.

$$k = D/6 = 400/6 = 66.7 \text{ mm above and below the C.G.}$$

$$e = 60 \text{ mm} < 66.7 \text{ mm} \rightarrow \text{inside kern} \rightarrow \text{no tension from } P \text{ alone} \checkmark$$

Check : stress at the top fibre = $P/A\,(1 - 6e/D) = 7.5\,(1 - 0.9) = +0.75\ N/mm^2$ ✓ compression.

7. Cable Profile and Cable Zone

Cable zone is the zone in the depth of the beam within which the tendon profile (centroid) can be placed such that stresses are within permissible limits at all sections. Beyond it, tension / over-compression results.

Limiting eccentricities – at a section with M_g (transfer) and M_T (service)

- **Transfer** : (i) bottom fibre compression ≤ f_{ci}; (ii) top fibre tension ≥ f_{ti} (0 for Type 1).
- **Service** : (iii) top fibre compression ≤ f_{cw}; (iv) bottom fibre tension ≥ f_{tw} (0 for Type 1).

$$(1)\ e \le Z_b(f_{ct} - P_i/A)/P_i + M_g/P_i$$

$$(2)\ e \le Z_t(P_i/A - f_{ti})/P_i + M_g/P_i$$

$$(3)\ e \ge M_T/(\eta P_i) - Z_t(f_{cw} - \eta P_i/A)/(\eta P_i)$$

$$(4)\ e \ge M_T/(\eta P_i) + Z_b(f_{tw} - \eta P_i/A)/(\eta P_i)$$

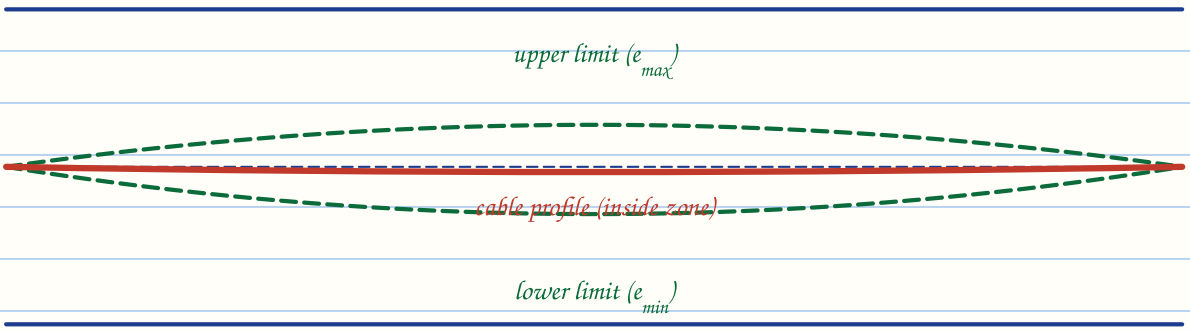


Fig. Cable zone for a simply supported beam

Concordant cable : cable profile such that the tendon stays at the C-line (no secondary moments). In a simply supported beam **any cable is concordant**; in continuous beams only some profiles are (secondary moments arise otherwise).

8. Design of PSC Beam – Rectangular Section (IS 1343)

Procedure (Type 1 – no tension, simply supported beam)

1. Assume section depth $D \approx L/20 \ldots L/25$, estimate self-weight → M_g, then M_q and $M_T = M_g + M_q$.
2. Required **section modulus** :

$$Z_b \geq (M_T - \eta M_g)/(\eta f_{ct} - f_{tw})$$

- f_{ct} = permissible compression at the bottom at transfer ; f_{tw} = permissible tension at the bottom at service (0 for Type 1).
- Derivation : eliminate $P/A + Pe/Z_b$ from the bottom fibre equations at transfer and at service.
- Similarly for the top fibre : $Z_t \geq (M_T - \eta M_g)/(f_{cw} - \eta f_{tt})$.

1. Choose the section (b, D) with $Z \geq$ required Z; recompute M_g with the actual self-weight.
2. Find P_i and **e** : $P_i (1/A + e/Z_b) = M_T/(\eta Z_b)$ (for $f_{tw} = 0$) — choose e (practical : y_b – cover) to get P_i.
3. Check stresses at **transfer** and **service** at mid-span (top & bottom) and at the **support**.
4. Choose steel : $A_{ps} = P_i/f_{pi}$ ($f_{pi} \approx 0.6$–$0.7 f_{pk}$).
5. Check ultimate moment of resistance $M_u \geq 1.5 M_T$ (IS 1343 – load factor), shear, deflection, end block, etc.

Derivation of Z_b : Transfer : $P_i/A + P_i e/Z_b - M_g/Z_b = f_{ct}$. Service : $\eta(P_i/A + P_i e/Z_b) - M_T/Z_b = f_{tw}$. Substituting → $Z_b = (M_T - \eta M_g)/(\eta f_{ct} - f_{tw})$.

Design Example 5 – Rectangular beam

Design a simply supported post-tensioned beam of span 10 m for an imposed load of 10 kN/m. Type 1 (no tension) ; f_{ct} = 15 N/mm² at transfer (bottom), f_{cw} = 16 N/mm² at service (top), f_{tt} = f_{tw} = 0 ; η = 0.8 ; concrete M40, density 25 kN/m³.

Step 1 – Trial section 300 × 500

$$A = 1.5\times10^{5}\ mm^2\ ;\ Z = 300\times500^2/6 = 12.5\times10^{6}\ mm^3\ ;\ self\ weight = 3.75\ kN/m$$

$$M_g = 3.75\times10^2/8 = 46.9\ kN\cdot m\ ;\ M_q = 10\times10^2/8 = 125\ kN\cdot m\ ;\ M_T = 171.9\ kN\cdot m$$

$$Z_{req} = (171.9 - 0.8\times46.9)\times10^6/(0.8\times15 - 0) = 134.4\times10^6/12 = 11.2\times10^6 < 12.5\times10^6\ \checkmark\ OK$$

Step 2 – Prestressing force and eccentricity

$$P_i(1/A + e/Z) = M_T/(\eta Z) = 171.9/(0.8\times12.5) = 17.19\ N/mm^2$$

Adopt e = 120 mm (cover and duct limit; larger e gives top tension at transfer) :

$$1/A + e/Z = 6.667\times10^{-6} + 9.6\times10^{-6} = 1.627\times10^{-5} \rightarrow P_i = 17.19/1.627\times10^{-5} = 1057\ kN$$

Step 3 – Check stresses at mid-span (N/mm²)

Stage	Top fibre	Bottom fibre
Transfer (P_i, M_g)	7.05 − 10.15 + 3.75 = **+0.65** ✓	7.05 + 10.15 − 3.75 = **13.45** ≤ 15 ✓
Service (ηP_i, M_T)	0.8(7.05 − 10.15) + 13.75 = **11.27** ≤ 16 ✓	0.8(17.20) − 13.75 = **≈ 0** ✓

Step 4 – Steel

Initial stress ≈ 1070 N/mm² (0.6 f_{pk} for f_{pk} = 1770) → A_{ps} = 1057×10³/1070 = 988 mm² → **10 strands Ø 12.7 mm (987 mm²)**. Ducts : 1 cable of 10 strands (or 2 × 5).

Step 5 – Support section

At the support M = 0 → tendon should be at C.G. (e = 0) or within kern (e ≤ D/6 = 83 mm) : uniform P/A = 7.05 N/mm² ✓ . Use a **parabolic profile** from e = 120 mm at mid to e = 0 at supports (or raised cable).

> Final : 300 × 500 mm beam, P_i = 1057 kN, e = 120 mm at mid-span, 10 strands Ø 12.7 mm, stirrups by shear design (Unit III shear).

9. Design of I-Section Beam

For longer spans an I-section (symmetrical / unsymmetrical) is more efficient: the **flanges** carry the stresses, the thin web is used mainly for shear – high Z/A ratio → larger kern, smaller P.

Efficiency of the section : $k_t + k_b = (Z_t + Z_b)/A$ (larger = better)

Design Example 6

Design an I-section beam, simply supported, span 14 m, imposed load 12 kN/m. Trial section : flanges 400 × 100 each, web 100 × 600, D = 800 mm. Permissible : $f_{ct} = 15$, $f_{cw} = 16$, $f_{tw} = f_{tt} = 0$; $\eta = 0.8$.

$$A = 2\times400\times100 + 100\times600 = 1.4\times10^5 \text{ mm}^2$$

$$I = 400\times800^3/12 - 300\times600^3/12 = 1.7067\times10^{10} - 5.4\times10^9 = 1.1667\times10^{10} \text{ mm}^4$$

$$Z_t = Z_b = I/400 = 29.17\times10^6 \text{ mm}^3 \text{ ; self weight} = 0.14 \times 25 = 3.5 \text{ kN/m}$$

$$M_g = 3.5 \times 14^2/8 = 85.75 \text{ kN·m} ; M_q = 12 \times 14^2/8 = 294 \text{ kN·m} ; M_T = 379.75 \text{ kN·m}$$

$$Z_{req} = (379.75 - 0.8\times85.75)/(0.8\times15) = 311.15/12 = 25.9\times10^6 < 29.17\times10^6 \checkmark$$

Prestressing force & eccentricity

$$P_i(1/A + e/Z) = M_T/(\eta Z) = 379.75/(0.8 \times 29.17) = 16.27 \text{ N/mm}^2$$

Max practical $e = y_b - \text{cover}(150) = 400 - 150 = 250$ mm : $1/A + e/Z = 7.143\times10^{-6} + 8.57\times10^{-6} = 1.571\times10^{-5}$

$$P_i = 16.27/1.571\times10^{-5} = 1036 \text{ kN}$$

Stage	Top fibre (N/mm²)	Bottom fibre (N/mm²)
Transfer	7.40 − 8.88 + 2.94 = **+1.46** ✓	7.40 + 8.88 − 2.94 = **13.34** ≤ 15 ✓
Service	0.8(7.40−8.88) + 13.02 = **11.84** ≤ 16 ✓	0.8(16.28) − 13.02 = **≈ 0** ✓

Steel : $A_{ps} = 1036\times10^3/1100 = 942$ mm² → **10 strands Ø 12.7** (987 mm²). Profile : parabolic, e = 250 mm at mid-span reducing to 0 – 100 mm at supports (within kern $k = Z/A = 208$ mm).

10. Analysis of Shear in PSC Beams

Shear stress in PSC beams must be checked for **principal tension**, which causes diagonal cracking. Prestress (axial compression) reduces the principal tension.

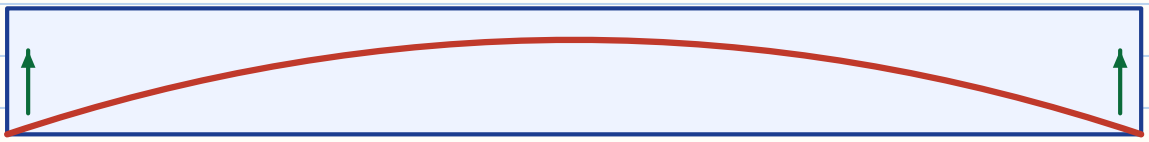
Principal stresses

$$\sigma_{1,2} = f/2 \pm \sqrt{[(f/2)^2 + \tau^2]} \quad \tan 2\alpha = 2\tau / f$$

- f = longitudinal compressive stress ($P/A \pm Pe\,y/I \pm M\,y/I$) ; τ = shear stress = $V\,S/(I\,b)$ (neglecting vertical stress).
- Principal tension $\sigma_t = -\sigma_2 = \sqrt{[(f/2)^2 + \tau^2]} - f/2$. Without prestress ($f = 0$) $\sigma_t = \tau \rightarrow$ with prestress it is less.
- Cracking when σ_t reaches tensile strength $f_t = 0.24\sqrt{f_{ck}}$ (direct tension).
- Maximum principal tension is normally at the **junction of web and flange** or at the C.G. of I-section.

Effect of tendon curvature

- Inclined (harped / parabolic) tendon gives a vertical component $V_p = P \sin\theta$ opposing the external shear.
- **Net shear** $V_{net} = V - V_p$; shear stress $\tau = (V - P \sin\theta)\,S/(I\,b)$.



$V_p = P \sin\theta$ opposes V

Fig. Tendon curvature reduces net shear

Types of cracks : (i) **flexure-shear cracks** – flexural crack propagates diagonally (cracked section, V_{cr}); (ii) **web-shear cracks** – principal tension in web exceeds f_t before flexure cracks (uncracked, V_{co}), common near supports and in thin webs.

IS 1343 gives shear resistance in two cases : V_{co} (un-cracked in flexure) and V_{cr} (cracked in flexure). The **lesser** controls.

11. Shear Design as per IS 1343

(a) Section uncracked in flexure (V_{co})

$$V_{co} = 0.67\, b\, D \sqrt{(f_t^2 + 0.8\, f_{cp} f_t)} \; (+ V_p)$$

- $f_t = 0.24 \sqrt{f_{ck}}$ (max principal tensile stress) ; f_{cp} = compressive stress at centroidal axis due to prestress (P_e/A) ; b = width of web ; D = overall depth.
- Check where there is no flexural tension in the section at ultimate (e.g., near supports).

(b) Section cracked in flexure (V_{cr})

$$V_{cr} = 0.037\, b\, d \sqrt{f_{ck}} + M_0\, V / M \geq 0.1\, b\, d \sqrt{f_{ck}}$$

- M_0 = moment necessary to produce zero stress at the extreme tension fibre = $0.8\, f_{pt}\, I / y_t$; f_{pt} = stress at tension fibre due to prestress only ; V, M = shear and moment at ultimate at the section ; d = effective depth.
- V_{cr} is calculated at sections more than $D/2$ from the support, and V_c = lesser of V_{co} and V_{cr} (where both apply).

Shear reinforcement

- If $V \leq V_c$ provide only **minimum (nominal) shear reinforcement**; if $V > V_c$ design stirrups for $(V - V_c)$.
- Vertical stirrups :

$$A_{sv} / s_v = (V - V_c) / (0.87\, f_y\, d_t)$$

- Minimum : $A_{sv} / (b\, s_v) \geq 0.4 / (0.87\, f_y)$ (IS 1343 cl. 22.4.2).
- Max spacing $s_v \leq 0.75\, d_t$ (or 4 b) and ≤ 450 mm.
- d_t = depth to the centroid of the stirrup – i.e. distance from extreme compression fibre to the longitudinal tension steel.
- Shear stress is limited to $\tau_{c,max}$ (IS 1343 Table 6) – otherwise enlarge the web.

Example 7 – Shear (uncracked)

The beam of Design Example 5 (300 × 500, M40, $P_e = 0.8 \times 1057 = 845$ kN, tendon at C.G. at the supports). Total service load = 13.75 kN/m, span 10 m. Check shear at the support section.

$$V_u = 1.5 \times 13.75 \times 10/2 = 103 \text{ kN}; \; f_{cp} = 845\times10^3/1.5\times10^5 = 5.63 \text{ N/mm}^2$$

$$f_t = 0.24\sqrt{40} = 1.52 \text{ N/mm}^2$$

$$V_{co} = 0.67 \times 300 \times 500 \times \sqrt{(1.52^2 + 0.8\times5.63\times1.52)} = 100500 \times \sqrt{(2.31 + 6.85)} = 100500 \times 3.03$$

$$= 304 \text{ kN}$$

$V_u = 103$ kN < $V_{co} = 304$ kN → only **minimum shear reinforcement** is needed : $A_{sv}/(b\, s_v) \geq 0.4/(0.87 \times 415) = 0.00111$. 2-legged 8 mm stirrups (101 mm²): $s_v = 101/(0.00111 \times 300) = 303$ mm → limited to 0.75 $d_t \approx 280$ → **provide 8 mm @ 250 mm c/c**.

Example 8 – Shear (stirrups required)

A beam web $b = 300$, $D = 500$, $d_t = 400$, $V_c = 304$ kN (above), ultimate shear $V = 450$ kN at a section. $f_y = 415$ N/mm². Design stirrups.

$$A_{sv}/s_v = (450 - 304)\times10^3 / (0.87 \times 415 \times 400) = 146000/144420 = 1.01 \text{ mm}^2/\text{mm}$$

2-legged 10 mm : $A_{sv} = 157$ mm² → $s_v = 157/1.01 = 155$ mm → **provide 10 mm @ 150 mm c/c**

Check $s_v \leq 0.75 \times 400 = 300$ mm ✓ ; minimum $A_{sv}/s_v = 0.00111 \times 300 = 0.33$ mm²/mm ✓ (1.01 > 0.33).

Example 9 – V_{cr} (cracked section)

At a section : $M = 150$ kN·m, $V = 120$ kN (ultimate), $b = 300$, $d = 370$, $f_{ck} = 40$, $f_{pt} = 8$ N/mm² (prestress tension fibre = bottom), $I = 3.125 \times 10^9$, $y_t = 250$.

$$M_0 = 0.8 \times 8 \times 3.125\times10^9/250 = 80 \text{ kN·m}$$

$$V_{cr} = 0.037 \times 300 \times 370 \times \sqrt{40} + 80 \times 120/150 = 26.0 + 64.0 = 90 \text{ kN}$$

Minimum : $0.1 \times 300 \times 370 \times 6.32 = 70.2$ kN ✓ . Since $V = 120 > V_{cr} = 90$ kN, stirrups for 30 kN : $A_{sv}/s_v = 30000/(0.87\times415\times370) = 0.22$ → provide minimum (0.33).

> Take V_c as the **lesser** of V_{co} and V_{cr} and always provide minimum stirrups.

12. Transmission Length and End Block (Introduction)

Transmission length (pre-tensioned members)

- In pre-tensioned members the stress is transferred from tendon to concrete **by bond**. At the free end the steel stress is zero; it builds up to full value over a length called the **transmission length L_t**.
- Within L_t the prestress is less than the effective value → the shear & flexure design near supports must consider reduced prestress.
- Factors affecting L_t : diameter & type of tendon (strands, indented wires), concrete strength at transfer, surface condition, rate of release (gradual – sudden increases L_t), Hoyer effect, tendon stress.
- IS 1343 : $L_t \approx 30\,\varphi - 100\,\varphi$ depending on the type; some practical values: plain wires ≈ 100 φ ; strands ≈ 30–60 φ; (φ = diameter).

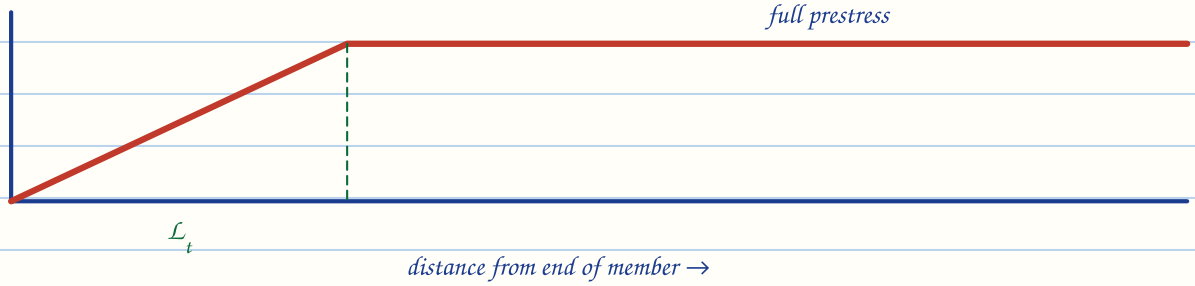


Fig. Variation of steel stress over the transmission length

End block (post-tensioned members)

- At the end of a post-tensioned member the anchorage force P is applied as a **concentrated load** on a small area. The stress spreads over the full section only after a distance ≈ depth of the member – this length is the **end block** (anchorage zone).
- Within the end block high **transverse tensile stresses** arise : **bursting tension** (centre of block, along load axis) and **spalling tension** (at the loaded face corners/edges); also high bearing stresses under the plate.
- Reinforcement : closely spaced **helices / stirrups / mesh** (bursting steel) designed by Guyon / Magnel methods; bearing plates, local enlargement of the end. (Design not required in this syllabus.)

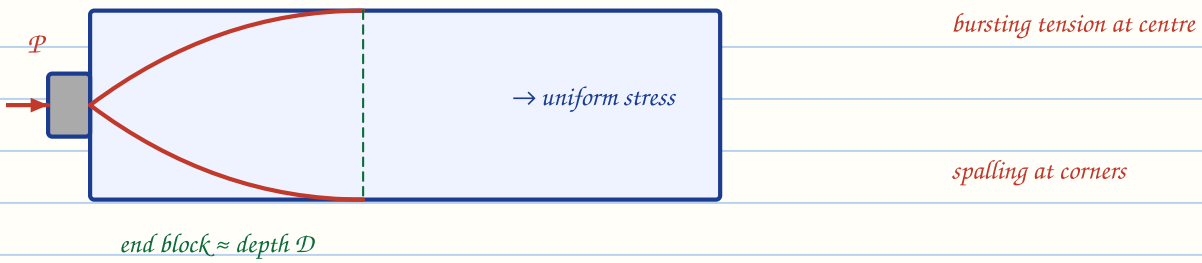


Fig. Anchorage zone (end block) – spread of prestress

13. Important Questions & Formula Sheet

Short answers

1. State the assumptions in the analysis of PSC section.
2. What is the kern ? Locate kern points for a rectangular section.
3. What is cable zone ? What is a concordant cable ?
4. Write the equivalent load for a parabolic tendon.
5. What is transmission length ? Define end block.
6. Why does prestress reduce principal tension ?

Long answers

1. A PSC beam 200×400 with a parabolic tendon … find stresses at mid-span and at supports (analysis).
2. Design a PSC rectangular beam for given span and load using IS 1343 (Z_b procedure).
3. Design an I-section PSC beam – check stress at transfer and service.
4. Explain the shear failure of PSC beams. Derive V_{co} and V_{cr} (IS 1343) and design shear reinforcement.
5. Explain the concept of load balancing; write about kern points and thrust line.

Formulae

Item	Formula
Stress	$f = P/A \pm Pe\,y/I \pm M\,y/I$
Kern	$k_t = Z_b/A$; $k_b = Z_t/A$ ($D/6$ for rect.)
Parabolic equivalent UDL	$w_p = 8Pe/L^2$
Bent tendon	$W = 4Pe/L$; $V_p = P \sin\theta$
Design Z_b	$(M_T - \eta M_q)/(\eta f_{ct} - f_{tw})$
V_{co}	$0.67\,b\,D\sqrt{(f_t^2 + 0.8\,f_{cp}\,f_t)}$
V_{cr}	$0.037\,b\,d\sqrt{f_{ck}} + M_0 V/M$
Stirrups	$A_{sv}/s_v = (V - V_c)/(0.87\,f_y\,d_t)$

Note : Permissible stress values and shear coefficients should be taken from IS 1343 (Tables 5 & 6, cl. 22) as given in your question paper / data book.